

**COURSE: DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT**

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The Scenario:

A local library currently tracks its books and members in a physical notebook. Books are being lost, and there is no way to know who has which book. The goal is to build the SQL foundation for "BookFlow" so that every book has a unique ISBN and no member can sign up without a valid (unique) email. The requirements are:

- Create a database named bookflow_db.
- Create a books table (book_id, title, isbn UNIQUE, published_year) and a members table (member_id, full_name, email UNIQUE).
- Enforce PRIMARY KEY on both IDs, NOT NULL on title, and a CHECK constraint so published_year is not in the future (< 2027).
- Seed 3 books and 3 members, then test that the rules actually reject bad data.

Screenshots of the Scenario:

1)Creating the database and books table:

```
Microsoft Windows [Version 10.0.26200.8875]
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C:\Users\rohit>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 20
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE IF NOT EXISTS bookflow_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE books(
  -> book_id INT AUTO_INCREMENT PRIMARY KEY,
  -> title VARCHAR(225) NOT NULL,
  -> isbn VARCHAR(13) NOT NULL UNIQUE,
  -> published_year INT,
  -> CONSTRAINT chk_published_year
  -> CHECK (published_year < 2027)
  -> );
Query OK, 0 rows affected (0.05 sec)
```

2)Creating table for table “members”:

```
mysql> CREATE TABLE members(
  -> member_id INT AUTO_INCREMENT PRIMARY KEY,
  -> full_name VARCHAR(100) NOT NULL,
  -> email VARCHAR(150) NOT NULL UNIQUE
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO books(title,isbn,published_year)
  -> VALUES
  -> ('The Alchemist','9780061122415',1988),
  -> ('Clean Code','9780132350884',2008),
  -> ('Atomic Habits','9780735211292',2018);
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

3) Inserting values of books and members

```
mysql> INSERT INTO books(title,isbn,published_year)
-> VALUES
-> ('The Alchemist','9780061122415',1988),
-> ('Clean Code','9780132350884',2008),
-> ('Atomic Habits','9780735211292',2018);
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO members(full_name,email)
-> VALUES
-> ('Anil Kumar','anil.kumar@example.com'),
-> ('Priya Sharma','priya.sharma@example.com'),
-> ('Ravi Verma','ravi.verma@example.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Verification: Inspect the table structures

```
mysql> SELECT*FROM books;
+-----+-----+-----+-----+
| book_id | title       | isbn       | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist | 9780061122415 | 1988          |
| 2       | Clean Code    | 9780132350884 | 2008          |
| 3       | Atomic Habits | 9780735211292 | 2018          |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT*FROM members;
+-----+-----+-----+
| member_id | full_name | email |
+-----+-----+-----+
| 1         | Anil Kumar | anil.kumar@example.com |
| 2         | Priya Sharma | priya.sharma@example.com |
| 3         | Ravi Verma | ravi.verma@example.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> DESCRIBE books;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id    | int       | NO   | PRI | NULL    | auto_increment |
| title      | varchar(225) | NO   |     | NULL    |                |
| isbn       | varchar(13) | NO   | UNI | NULL    |                |
| published_year | int       | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> DESCRIBE members;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| member_id  | int       | NO   | PRI | NULL    | auto_increment |
| full_name  | varchar(100) | NO   |     | NULL    |                |
| email      | varchar(150) | NO   | UNI | NULL    |                |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Test 1:

Duplicate ISBN (UNIQUE violation) INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000);

Test 2:

NULL title (NOT NULL violation) INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010);

Test 3:

Future publication year (CHECK violation) INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888888', 2030);

Test 4:

Duplicate email (UNIQUE violation) INSERT INTO members (full_name, email) VALUES ('Anil Clone', 'anil.kumar@example.com');

Output of Test1, Test2, Test3, Test4:

```
mysql> INSERT INTO books(title,isbn,published_year)
-> VALUES('Fake Copy','9780061122415',2000);
ERROR 1062 (23000): Duplicate entry '9780061122415' for key 'books.isbn'
mysql> INSERT INTO books(title,isbn,published_year)
-> VALUES(NULL,'9999999999999',2010);
ERROR 1048 (23000): Column 'title' cannot be null
mysql> ^C
-> \c
mysql> INSERT INTO books(title,isbn,published_year)
-> VALUES('Time Traveler','8888888888888',2030);
ERROR 3819 (HY000): Check constraint 'chk_published_year' is violated.
mysql> INSERT INTO members(full_name,email)
-> VALUES('Anil Clone','anil.kumar@example.com');
ERROR 1062 (23000): Duplicate entry 'anil.kumar@example.com' for key 'members.email'
mysql> |
```